Risk Alert Classifier — Theory Notes

Short definitions for the supervised learning project

1. Logistic Regression

A supervised classification algorithm that estimates the probability of a class. In this project, class 1 represents a high-risk customer and class 0 represents a low-risk customer.

2. Accuracy and imbalanced data

Accuracy is the proportion of all predictions that are correct. It can be misleading when one class is much larger than the other, because a model may predict the majority class and still appear accurate.

3. Type-I and Type-II errors

Type-I error (false positive): a low-risk customer is flagged as high risk. Type-II error (false negative): a high-risk customer is incorrectly labelled low risk.

4. Precision

Of all customers predicted as high risk, precision is the proportion who are actually high risk.

5. Recall / True Positive Rate (TPR)

Of all truly high-risk customers, recall is the proportion correctly identified. Higher recall means fewer high-risk customers are missed.

6. F1-score

The harmonic mean of precision and recall. It is useful when we want a single score that balances both metrics.

7. False Positive Rate (FPR)

The proportion of truly low-risk customers who are incorrectly flagged as high risk.

8. ROC curve and ROC-AUC

The ROC curve plots TPR against FPR at different classification thresholds. ROC-AUC summarizes the model's ability to rank high-risk customers above low-risk customers; higher is generally better.

9. Class imbalance

Class imbalance means one target class has many more examples than another. Models may learn the majority class better and miss the minority class.

10. KNN Imputer

K-Nearest Neighbors Imputer fills missing numeric values using values from similar rows. It should be fitted on training data only to avoid leaking information from the test set.

11. Under-sampling and over-sampling

Under-sampling removes examples from the majority class. Over-sampling increases the number of minority-class examples, commonly by repeating existing examples.

12. SMOTE and ADASYN

SMOTE creates synthetic minority-class examples between nearby examples. ADASYN creates more synthetic examples in areas where minority examples are harder to learn.

13. Decision Tree and Random Forest

A Decision Tree learns a sequence of feature-based rules. A Random Forest combines many trees and often generalizes better than a single tree.

14. RandomizedSearchCV and GridSearchCV

RandomizedSearchCV tests a sample of parameter combinations. GridSearchCV evaluates each specified combination in a small grid. Cross-validation estimates performance using training data.

15. Business interpretation

False negatives may allow risky customers to go unnoticed. False positives can create unnecessary reviews for low-risk customers. A final model should balance these costs and be validated before real deployment.